



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL ASSESSMENT MARKING GUIDELINE 2023 TECHNOLOGY GRADE 9

SECTION A

Question 1

- One mark per answer.
- There are no half marks.

No.	Expected answers		
1.1	A	constraint	✓
1.2	C	communicate the model	✓
1.3	B	make	✓
1.4	C	budget	✓
1.5	D	orthographic	✓
1.6	A	specifications	✓
1.7	B	isometric	✓
1.8	C	size.	✓
1.9	B	hidden detail line	✓
1.10	D	evaluation	✓
1.11	C	millimetres	✓
1.12	C	scale	✓
1.13	A	shading	✓
1.14	B	flow diagram	✓
1.15	B	design brief	✓
1.16	C	initial drawings	✓
1.17	A	texture, colour, shadow and shading	✓
1.18	C	ergonomics	✓
1.19	D	free-hand drawing	✓
1.20	A	Identifying material—collecting material—measuring material—cutting— assembling—testing	✓
1.21	C	use materials economically.	✓
1.22	B	The object is three times bigger than the drawing.	✓
1.23	A	The working drawings of selected idea	✓
1.24	D	evaluate and select the best idea	✓

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Question 2

- There are no half marks.

No.	Expected answer	Clarification	Mark
2.1	Mercy is frightened to enter her home in the dark because her light is off when she comes back from work ✓ OR Mercy is frightened to enter her home because of the darkness outside her house.	One mark for the correct answer	1
		No mark if the answer is not relevant to the identified problem in the scenario	
2.2	Design an electronic device ✓ that will detect the change in light intensity ✓ and automatically switches on the outside light. ✓	1 mark: mentioning an electronic circuit device 1 mark: reason for the design 1 mark: the function of the design	3
		2 marks: if only two of the above-mentioned aspects were addressed	
		1 mark: if only one of the above-mentioned aspects was addressed	
		No mark for a response that is not mentioned in the scenario	
2.3	The device should allow the LDR to detect the change in light intensity. ✓	1 mark for the correct answer	1
		No mark for an incorrect specification	
2.4	The device needs to be installed within 2 days. ✓	1 mark for the correct answer	1
		No mark for an incorrect constraint	

No.	Expected answer	Clarification	Mark
2.5	9V	Components - All six electronic components are correctly included in the circuit. - LDR ✓ - LED ✓ - Transistor ✓ - 470 Ω resistor ✓ - 1KΩ resistor ✓ - 9 V Battery ✓ Connection Correct connection of the LED in terms of: - Polarity ✓ - Correct connection of the transistor ✓ Positioning - Correct positioning of the 1 KΩ resistor ✓ - Correct positioning of the 470 Ω resistor ✓	10

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TOTAL FOR SECTION A:40

SECTION B

Question 3

- One mark per answer.
- There are no half marks.

No.	Expected answer		
3.1	B	SWITCH	✓
3.2	C	It acts as a switch and amplifies current.	✓
3.3	A	stores and releases electrical energy	✓
3.4	B	4 A	✓
3.5	D	440 Ω	✓
3.6	B	When the voltage increases, the current increases.	✓
3.7	A	When the LDR is exposed to a high light intensity, the resistance value decreases.	✓
3.8	C	330 $\Omega \pm 5\%$	✓
3.9	D	violet, green, red, silver	✓
3.10	C	Lamp 2 and Lamp 3 will stop working.	✓
3.11	B	When the switch is thrown at A, light bulb 1 will glow.	✓
3.12	A	It emits light when current passes through it.	✓
3.13	C	The hand movement is detected by the sensor and closes the circuit allowing warm air to blow on the hands, drying them.	✓
3.14	A	the use of electricity to coat metal with another metal	✓
3.15	D	Zinc	✓
3.16	B	painting	✓
3.17	B	Corrosion is the deterioration of metal after it is exposed to oxygen and water.	✓
3.18	D	Copper sulphate	✓
3.19	C	storing grain	✓
3.20	A	Polyethylene Terephthalate	✓
3.21	C		✓
3.22	C	It is resistant to chemicals, tough, heat-resistant and a barrier to moisture.	✓
3.23	B	It is a process of making new products from old materials.	✓
3.24	D	Collection—sorting—shredding—cleaning—melting—making of pellets	✓

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SECTION C

Question 4

- One mark per answer.
- There are no half marks.

No.	Expected answer	Clarification	Mark
4.1.	Drying ✓, Pickling ✓, Salting ✓	3 marks for 3 correct methods mentioned	3
		2 marks if only 2 correct methods were mentioned	
		1 mark if only 1 correct method was mentioned	
		No mark if no correct method was mentioned	
4.2	4.2.1 C✓ 4.2.2 A✓ 4.2.3 B✓	3 marks for 3 correctly matched items	3
		2 marks for 2 correctly matched items	
		1 mark for 1 correctly matched item	
		No mark if none of the items were correctly matched	
4.3	a. <u>Advantages</u> It removes the moisture from food ✓ Make an unfavourable environment for micro-organisms to grow ✓	2 marks for 2 correctly mentioned advantages	2
		1 mark for only 1 correctly mentioned advantage	
		No mark for incorrect advantages	
	b. <u>Disadvantages</u> Using too much salt is not good for our health ✓ Food may lose its nutritional value or quality ✓	2 marks for 2 correctly mentioned disadvantages	2
		1 mark for only 1 correctly mentioned disadvantage	
		No mark for incorrect disadvantages	

No.	Expected answer	Clarification	Mark
4.4	There will be food shortages. ✓ There will be spoilage of food. ✓	2 marks for 2 correctly mentioned negative impacts.	2
		1 mark for 1 correctly mentioned negative impact.	
		No mark for incorrect impacts.	
4.5	a. <u>Economy</u> - It will create job opportunities. ✓	1 mark for 1 relevant point on the importance of preserving food (economy).	1
		No mark for an irrelevant point.	
	b. <u>Society</u> - Food will be available throughout the year, even if it is out of season. ✓	1 mark for 1 relevant point on the importance of preserving food (society).	1
		No mark for an irrelevant point.	
4.6	It kills the micro-organisms. ✓ OR It prevents bacteria from growing. OR It removes moisture from food, preventing chemical reactions.	1 mark for the 1 correct reason	1
		No mark for an incorrect reason	
4.7	Good hygiene is important when preserving food because food that is not cooked, stored or handled correctly can cause food poisoning and other conditions. ✓	1 mark for 1 correct conclusion on the importance of good hygiene.	1
		No mark for an incorrect conclusion on the importance of good hygiene.	

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